



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 9: Consumer Theory VII

Patricia Galilea A.

ICS2523 Microeconomics

Lecture 9

Last lecture we

- Learnt about the Individual Demand
- Remembered about Demand Elasticities
- Learnt about the Market Demand

In today's lecture, we will learn about **Consumer Welfare**

- Compensated Variation
- Equivalent Variation
- Consumer Surplus

Consumer Welfare

Consumer Theory

13. Consumer Welfare

When the economic environment changes, a consumer may be made better off or worse off.

How can we measure in terms of money the gains and losses that individuals experience when prices change?

An essential problem in **welfare economics** is devising a monetary measure of the gains and losses that individuals experience when prices change.

Activity 1

How can we measure in terms of money the gains and losses that individuals experience when prices change?

Name a concept you learnt in Introduction to Economics to answer this question.



Wooclap.com **GDOFXU**

<https://app.wooclap.com/GDOFXU?from=instruction-slide>

13. Consumer Welfare

Consumer surplus is the classical measure of welfare change examined in elementary courses. However, it is an exact measure of welfare change only in special circumstances.

We will forget (for the moment) about consumer surplus, and we will study first two other definitions of welfare: **Compensated variation** and **Equivalent variation**.

13. Consumer Welfare

We will have an initial income (I_1) and a set of prices (P_1) under one policy regime and another income (I_2) and a set of prices (P_2) under another policy regime.

One way to **measure the welfare change** involved in moving from (I_1, P_1) to (I_2, P_2) is the difference in indirect utility $\rightarrow V(I_2, P_2) - V(I_1, P_1)$

If this difference is positive, then the policy change is worth doing (from the individual's point of view).

But, for some purposes, it is convenient to have a monetary measure of changes in consumer welfare....₇

13. Consumer's Welfare:

Compensated Variation (CV)

If, for some reason (e.g. particular public policy: tax, subsidy, new project), the prices of some good change from a set of prices (P_1) to a set of prices (P_2), an individual will have a lower level of utility (utility falls from U_1 to U_2)

The **Compensated Variation** is the amount of additional money an agent would need to reach its initial utility after a change in prices

$$\rightarrow CV = E(P_2, U_1) - E(P_1, U_1)$$

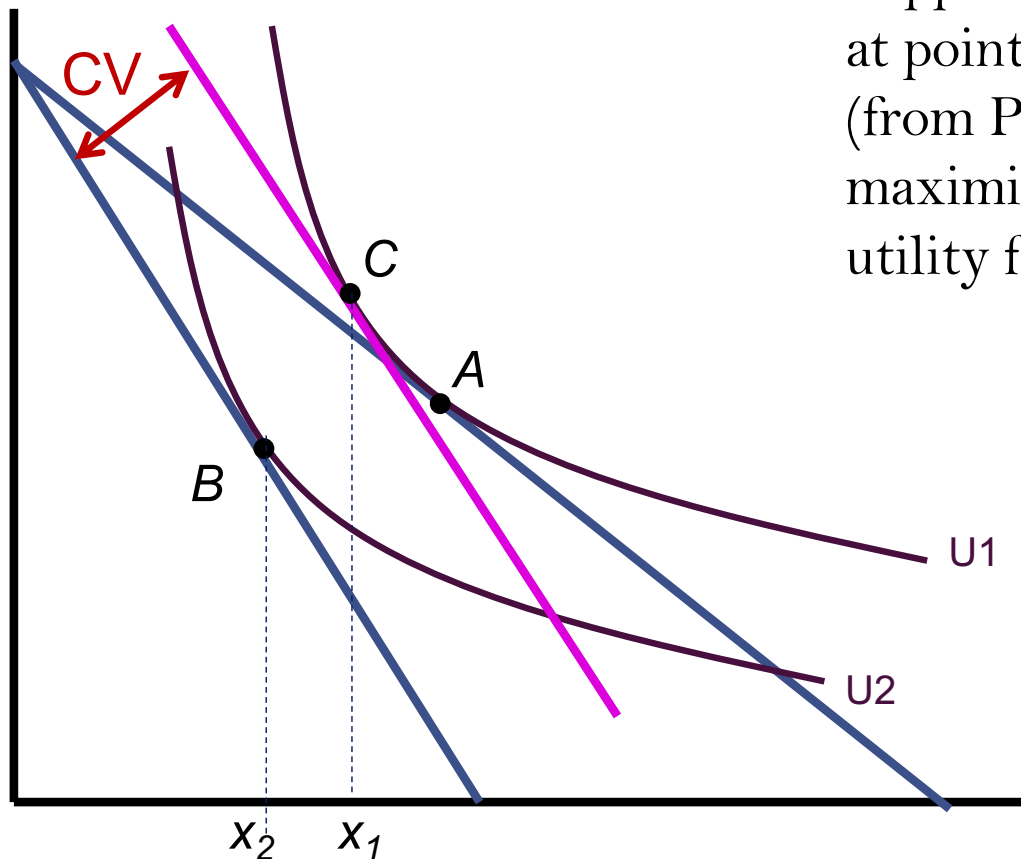
13. Consumer's Welfare:

Compensated Variation (CV)

$$CV = E(P_{x2}, P_y, U_1) - E(P_{x1}, P_y, U_1)$$

Quantity of y

$$P_{x1} < P_{x2}$$



Suppose the consumer is maximizing utility at point A. If the price of the good x rises (from P_{x1} to P_{x2}), the consumer will now maximize utility at point B $\rightarrow$ consumer's utility falls from U_1 to U_2

The consumer could be **compensated so that he can afford to remain on U_1**
 $\rightarrow$ CV is the amount that the individual would need to be compensated

13. Consumer's Welfare:

Compensated Variation (CV)

Compensating variation (CV): uses the new prices (P_{x2}) as the base and asks what income change would be necessary to compensate the consumer for the price change in terms of its impact on utility

$$U_1 = V(P_{x1}, P_y, I_1) = V(P_{x2}, P_y, I_1 + CV)$$

If we remember the expenditure function:

$$I_1 = E(P_{x1}, P_y, U_1)$$

$$I_1 + CV = E(P_{x2}, P_y, U_1)$$

$$\text{Therefore } \rightarrow CV = E(P_{x2}, P_y, U_1) - E(P_{x1}, P_y, U_1)$$

13. Consumer's Welfare:

Compensated Variation (CV)

For a constant level of utility (U_1), differential of $E(P_x, P_y, U_1)$ is:

$$dE(P_x, P_y, U_1) = \sum_i \frac{\partial E}{\partial P_i} dP_i$$

And if we integrate this equation we should obtain:

$$CV = E(P_{x2}, P_y, U_1) - E(P_{x1}, P_y, U_1)$$

Also, recalling identity $\frac{\partial E(P_x, P_y, U_1)}{\partial P_i} = h_i(P_x, P_y, U_1)$

We have that

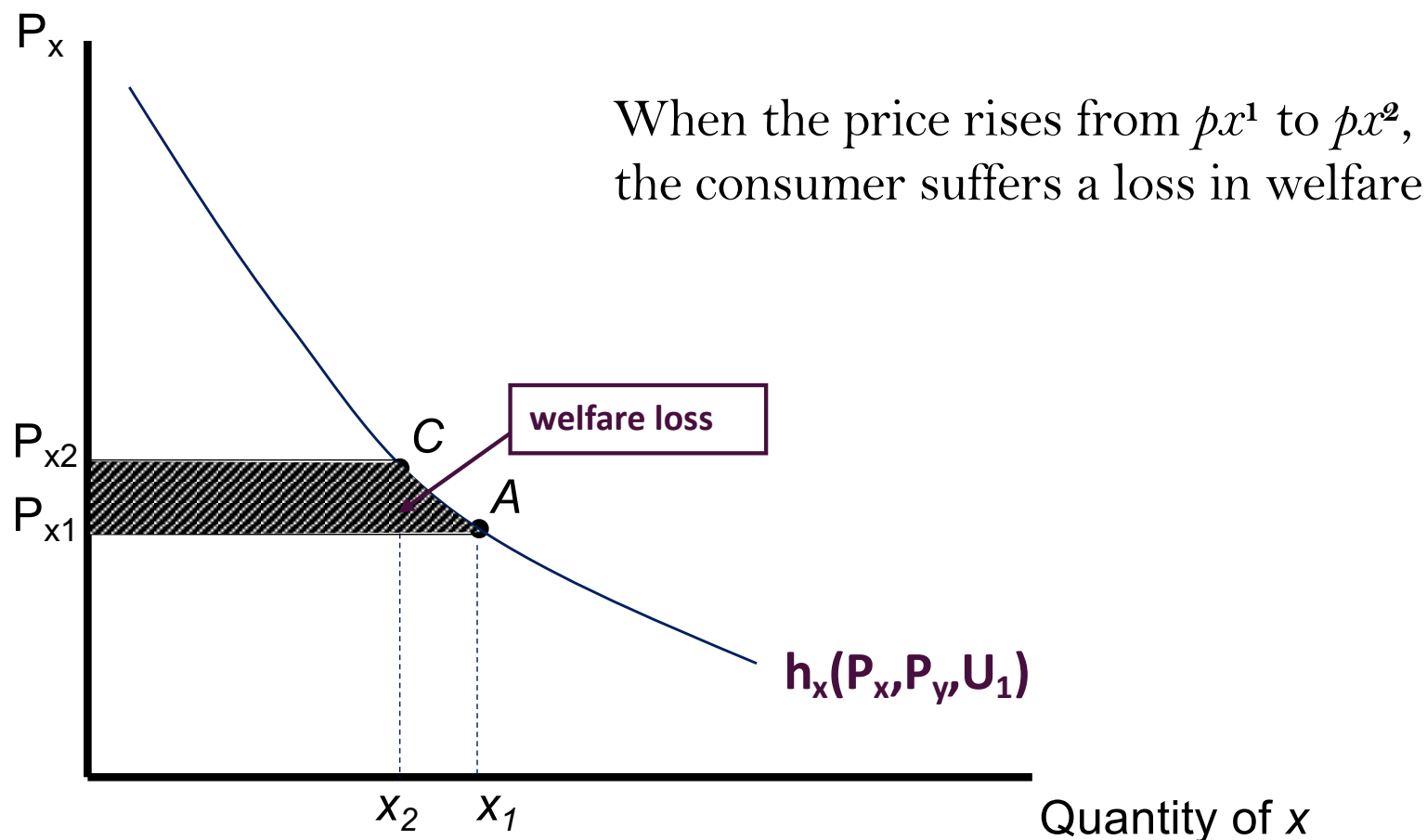
$$CV = \int_{P_{x1}}^{P_{x2}} \sum_i h_i(P_i, P_y, U_1) dP_i$$

→ CV can be interpreted graphically as the sum of the areas to the left of the Hicksian demand (with constant utility U_1)

13. Consumer's Welfare:

Compensated Variation (CV)

CV can also be represented by the shaded area below the compensated demand curve:



Example: **Compensated Variation**

Suppose a person's ranking of hamburgers (y) and soft drinks (x) could be represented by the utility function

$$U(x, y) = \sqrt{xy}$$

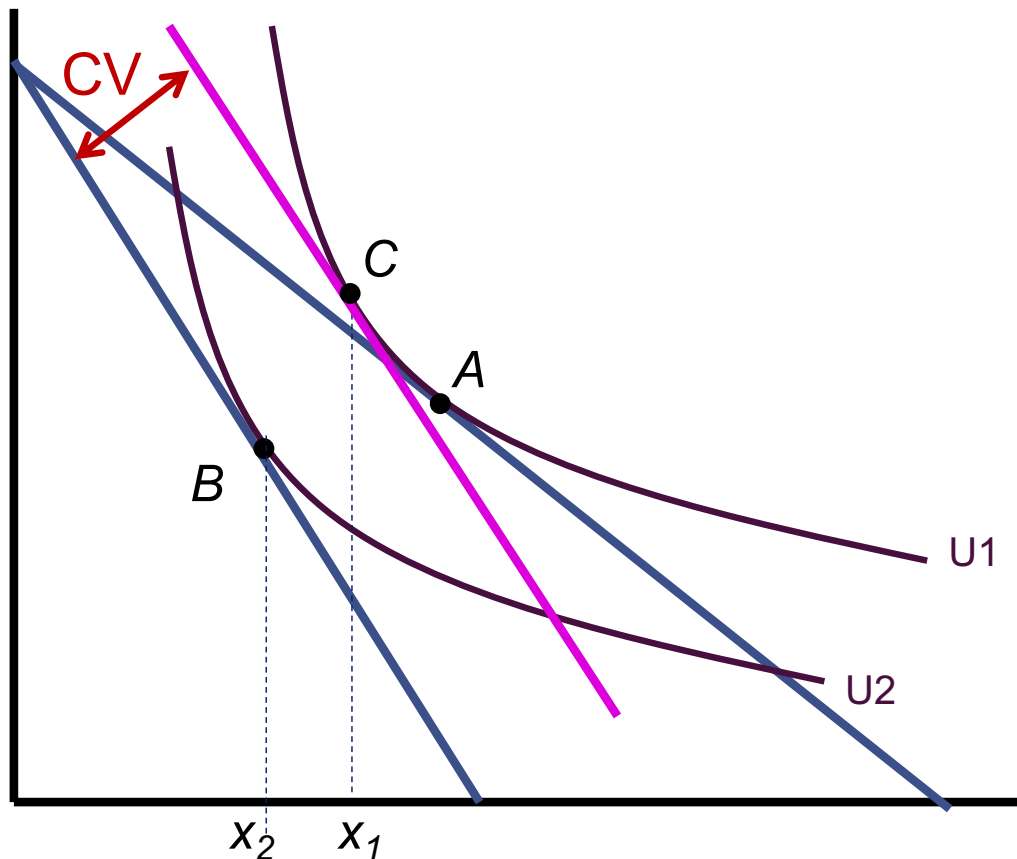
where the price of hamburgers is \$4 and soft drinks is \$1. This person's income is \$8.

Find the **Compensated Variation** if the price of soft drinks rises from \$1 to \$4

Example: Compensated Variation

Quantity of y

$$P_{x1} < P_{x2}$$



Point A maximizes utility at an income \$8 and $P_x = \$1$ and $P_y = \$4$.

Point B maximizes utility at an income \$8 and $P_x = \$4$ and $P_y = \$4$.

Point C doesn't maximize utility. It manages to reach the initial level of utility (U_1), with the new set of prices ($P_x = \$4$, $P_y = \$4$) and a different level of Income.

That difference in the levels of Income is CV

Quantity of x

Example: CV Long Solution

Point A maximizes utility at income \$8.

If we solve the maximization problem:

$$MRS = \frac{MU_x}{MU_y} = \frac{P_x}{P_y}$$
$$\frac{MU_x}{MU_y} = \frac{P_x}{P_y}$$

$$\frac{\frac{1}{2} y^{1/2}}{\frac{1}{2} x^{1/2}} = \frac{P_x}{P_y}$$
$$\frac{y}{x} = \frac{P_x}{P_y}$$
$$y = \frac{P_x}{P_y} x$$

Example: CV Long Solution

Replacing in the Budget Constraint:

$$\begin{aligned}P_x x + P_y y &= I \\P_x x + P_y \left(\frac{P_x}{P_y} x \right) &= I \\x_m^* &= \frac{I}{2P_x}, \\y_m^* &= \frac{I}{2P_y}\end{aligned}$$

In point A ($P_x=1$, $P_y=4$ and $I=8$)

$$\rightarrow x_m^* = \frac{8}{2 \cdot 1} = 4, y_m^* = \frac{8}{2 \cdot 4} = 1$$

The indirect utility at point A will be $V_A = \frac{I_A}{2} \frac{1}{P_x^{1/2} P_y^{1/2}} = 2$

Example: CV Long Solution

In point B ($P_x=4$, $P_y=4$ and $I=8$) Replacing in the Budget Restriction:

$$\rightarrow x_m^* = \frac{8}{2 \cdot 4} = 1, y_m^* = \frac{8}{2 \cdot 4} = 1$$

The indirect utility at point B will be $V_B = 1$

In point C ($P_x=4$, $P_y=4$ and $I=?$), but we know that has the same utility level as Point A:

$$\rightarrow V_C = \frac{I_C}{2} \frac{1}{P_x^{1/2} P_y^{1/2}}$$

$$\rightarrow 2 = \frac{I_C}{2} \frac{1}{2 \cdot 2} \rightarrow I_C = 16$$

Therefore, the CV should be $I_C - I_B = 16 - 8 = 8$

Example: CV Short Solution

If we realize then that we need to know the difference between the expenditure functions in points A and C, we only need to solve the dual problem (find Hicksian demand) and find the expenditure function:

The first-order condition is the same

$$y = \frac{P_x}{P_y}x$$

We replace it in the utility restriction:

$$\begin{aligned}\sqrt{xy} &= U \\ \sqrt{x \frac{P_x}{P_y}x} &= U\end{aligned}$$

$$x_h^* = U \sqrt{\frac{P_y}{P_x}}, y_h^* = U \sqrt{\frac{P_x}{P_y}}$$

Example: CV Short Solution

$$x_h^* = U \sqrt{\frac{P_y}{P_x}}, y_h^* = U \sqrt{\frac{P_x}{P_y}}$$

Then, the Expenditure Function will be:

$$E(U, P_x, P_y) = P_x x_h^* + P_y y_h^* = P_x U \sqrt{\frac{P_y}{P_x}} + P_y U \sqrt{\frac{P_x}{P_y}}$$
$$E(U, P_x, P_y) = 2U \sqrt{P_x P_y}$$

$$\text{At point A: } E(U, P_x, P_y) = 2 \cdot 2\sqrt{1 \cdot 4} = 8$$

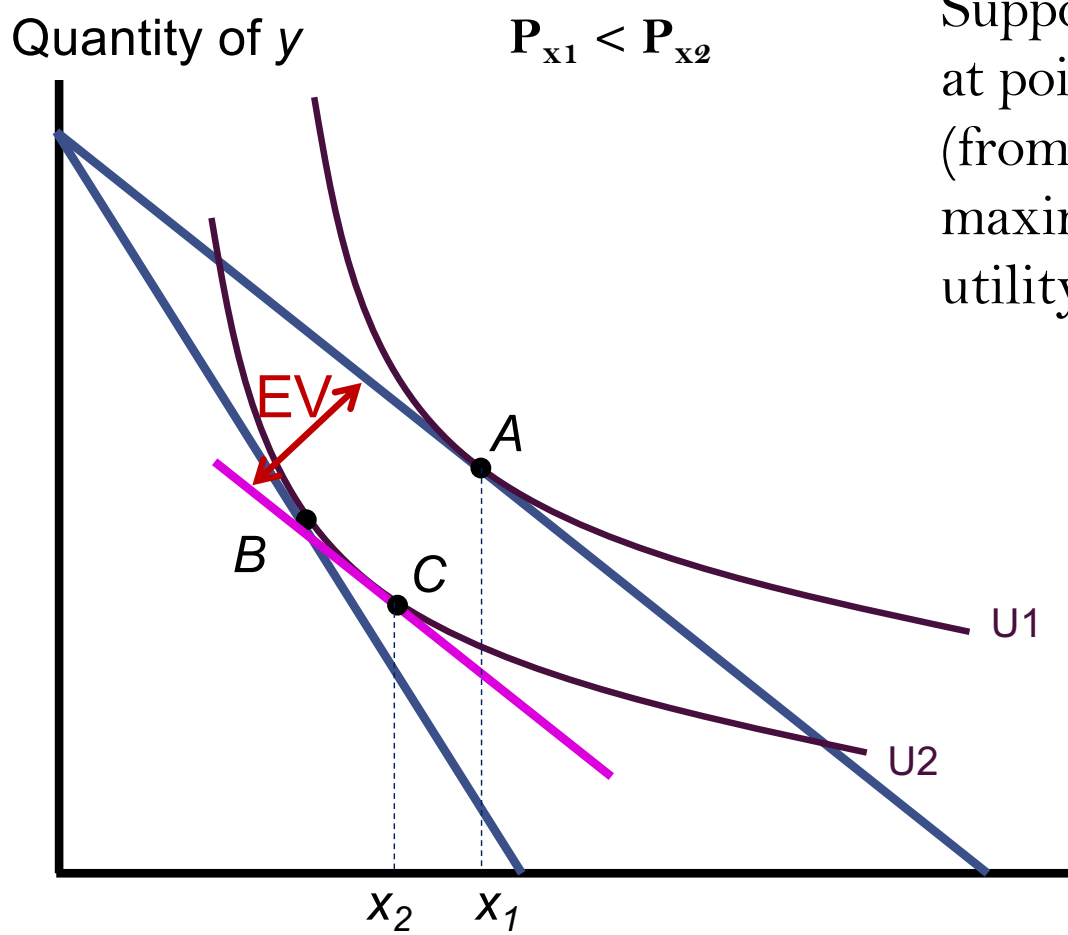
$$\text{At point C: } E(U, P_x, P_y) = 2 \cdot 2\sqrt{4 \cdot 4} = 16$$

Therefore, the CV is $16 - 8 = 8$

13. Consumer's Welfare:

Equivalent Variation (EV)

Amount of money a consumer would be willing to pay instead of the price change $\rightarrow EV = E(P_{x1}, P_y, U_2) - E(P_{x2}, P_y, U_2)$



Suppose the consumer is maximizing utility at point A. If the **price of the good x rises** (from P_{x1} to P_{x2}), the consumer will maximize utility at point B $\rightarrow$ consumer's utility falls from U_1 to U_2

The **consumer would be indifferent** between this price change or by **giving away EV** (in both situations she is in U_2) $\rightarrow$ EV is the amount that the individual would need to give away in order to be in an equivalent situation (U_2)

Quantity of x

13. Consumer's Welfare:

Equivalent Variation (EV)

Equivalent variation (EV): uses the current prices (P_1) as the base and asks what income change at the current prices would be equivalent to the proposed change of prices in terms of its impact on utility

$$U_2 = V(P_{x2}, P_y, I_1) = V(P_{x1}, P_y, I_1 + EV)$$

If we remember the expenditure function:

$$I_1 = E(P_{x2}, P_y, U_2)$$

$$I_1 + EV = E(P_{x1}, P_y, U_2)$$

$$\text{Therefore } \rightarrow EV = E(P_{x1}, P_y, U_2) - E(P_{x2}, P_y, U_2)$$

13. Consumer's Welfare:

Equivalent Variation (EV)

For a constant level of utility (U_2), differential of $E(P_x, P_y, U_2)$ is:

$$dE(P_x, P_y, U_2) = \sum_i \frac{\partial E}{\partial P_i} dP_i$$

And if we integrate this equation we should obtain:

$$EV = E(P_{x1}, P_y, U_2) - E(P_{x2}, P_y, U_2)$$

Recalling identity $\frac{\partial E(P_x, P_y, U_2)}{\partial P_i} = h_i(P_x, P_y, U_2)$

We have that

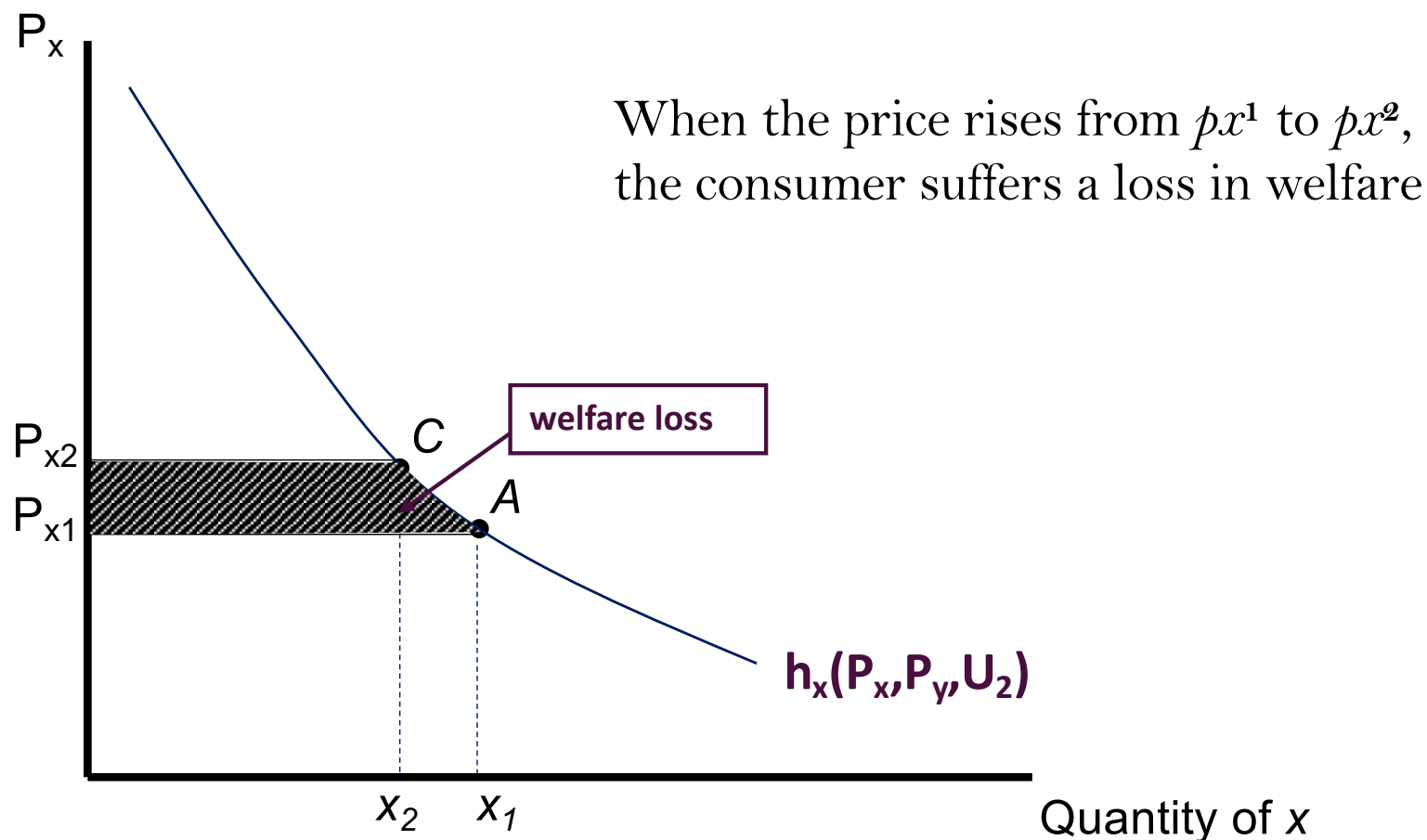
$$EV = - \int_{P_{x1}}^{P_{x2}} \sum_i h_i(P_i, P_y, U_2) dP_i$$

→ EV can be interpreted graphically as the sum of the areas to the left of the Hicksian demand (with constant utility)

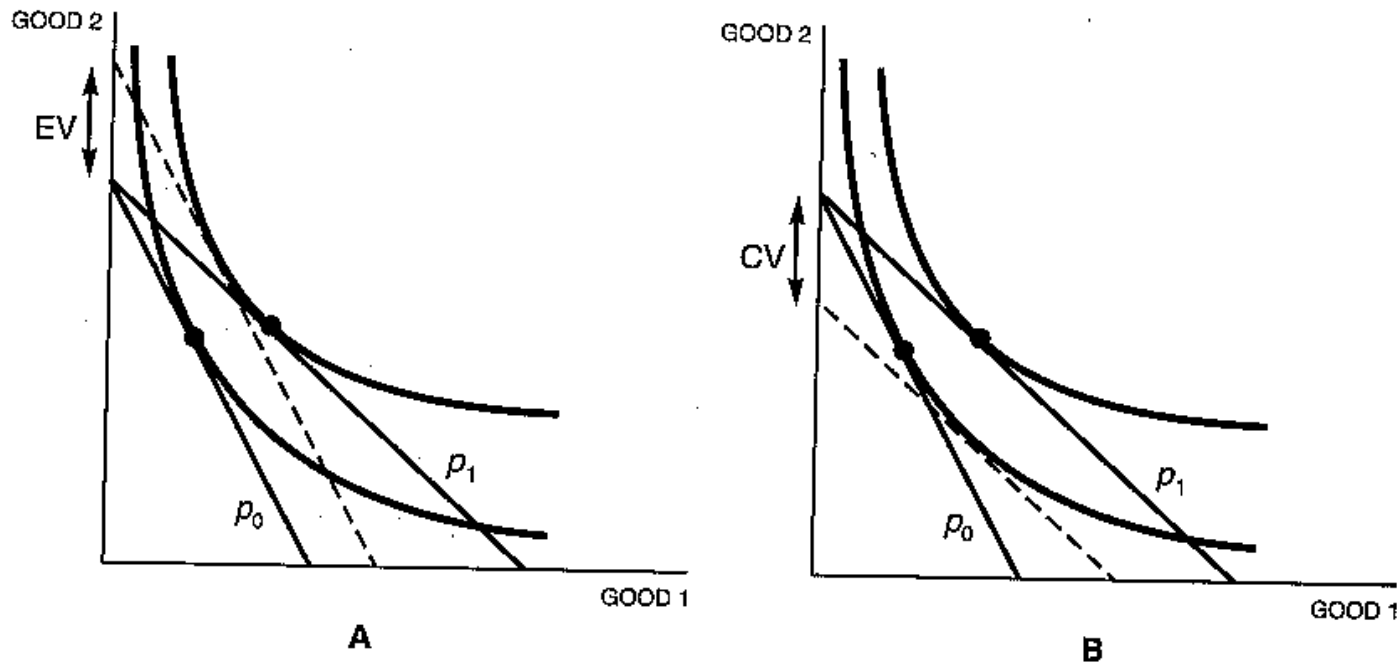
13. Consumer's Welfare:

Equivalent Variation (EV)

EV can also be represented by the shaded area below the compensated demand curve:



13. Consumer Welfare



Equivalent variation and compensating variation. In this diagram $p_2 = 1$ and the price of good 1 decreases from p_0 to p_1 . Panel A depicts the equivalent variation in income—how much additional money is needed at the original price p_0 to make the consumer as well off as she would be facing p_1 . Panel B depicts the compensating variation in income—how much money should be taken away from the consumer to leave him as well off as he was facing price p_0 .

Activity 2

Suppose a person's ranking of hamburgers (y) and soft drinks (x) could be represented by the utility function

$$U(x, y) = \sqrt{xy}$$

where the price of hamburgers is \$4 and soft drinks is \$1. This person's income is \$8.

Find the **Equivalent Variation** if the price of soft drinks rises from \$1 to \$4

13. Consumer's Welfare:

Consumer Surplus (CS)

Another way to look at this issue is to ask how much this person is willing to pay for the right to consume all of this good

Change in Consumer Surplus (CS): Measures the area to the left of the (Marshallian) demand curve between P_1 and P_2 .

$$CS = - \int_{P_1}^{P_2} x(t) dt$$

The area below the uncompensated demand curve and above the market price is called **consumer surplus** → the extra benefit the person receives by being able to make market transactions at the prevailing market price → **Marshallian demand curve is a Marginal Benefit Curve.**

13. Consumer's Welfare:

Consumer Surplus (CS)

Only when the consumer's preferences can be represented by a quasilinear utility function, CS is an exact measure of welfare change $\rightarrow$ in this case the magnitudes $CS=EV=CV$

Hint: Quasilinear: $U(x_1, x_2, x_3 \dots) = X_1 + U(X_2, x_3, x,4)$

Activity 3

Suppose a person's ranking of hamburgers (y) and soft drinks (x) could be represented by the utility function

$$U(x, y) = \sqrt{xy}$$

where the price of hamburgers is \$4 and soft drinks is \$1. This person's income is \$8.

Find the **Consumer Surplus** if the price of soft drinks rises from \$1 to \$4

Activity 3: Solution

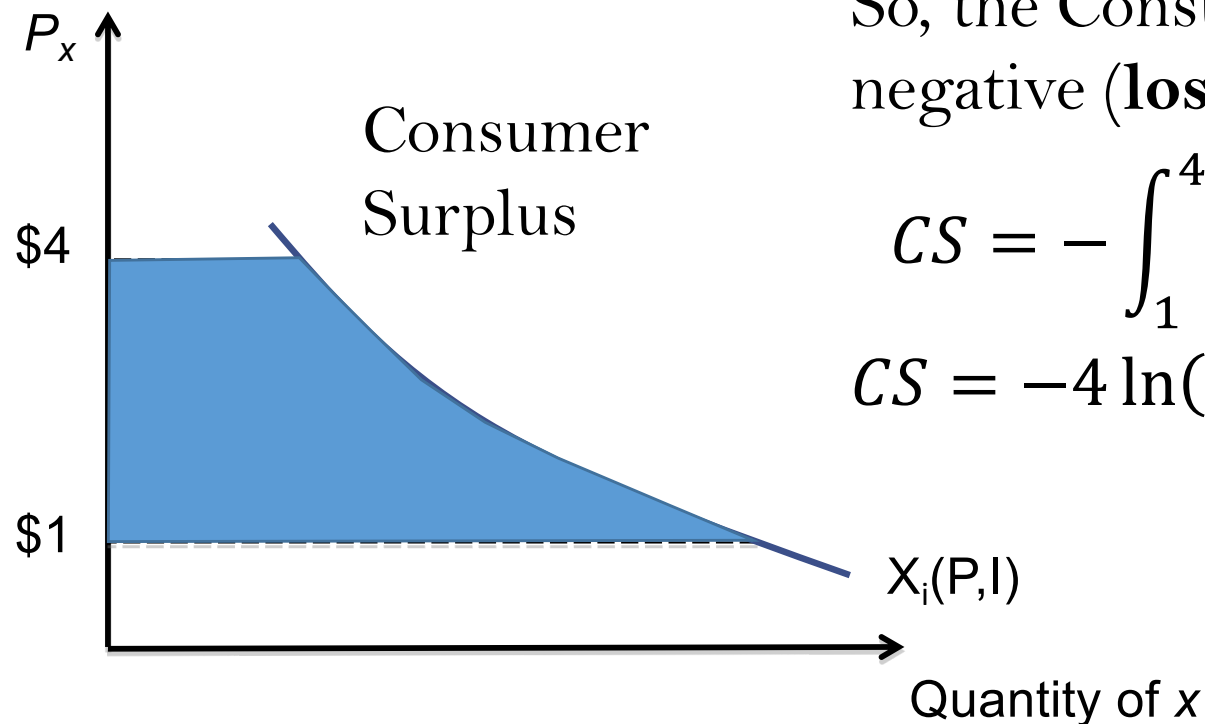
Recalling from last lecture, the Marshallian demand curve was:

$$x_m^* = \frac{I}{2P_x}$$

So, the Consumer Surplus will be negative (**loss**):

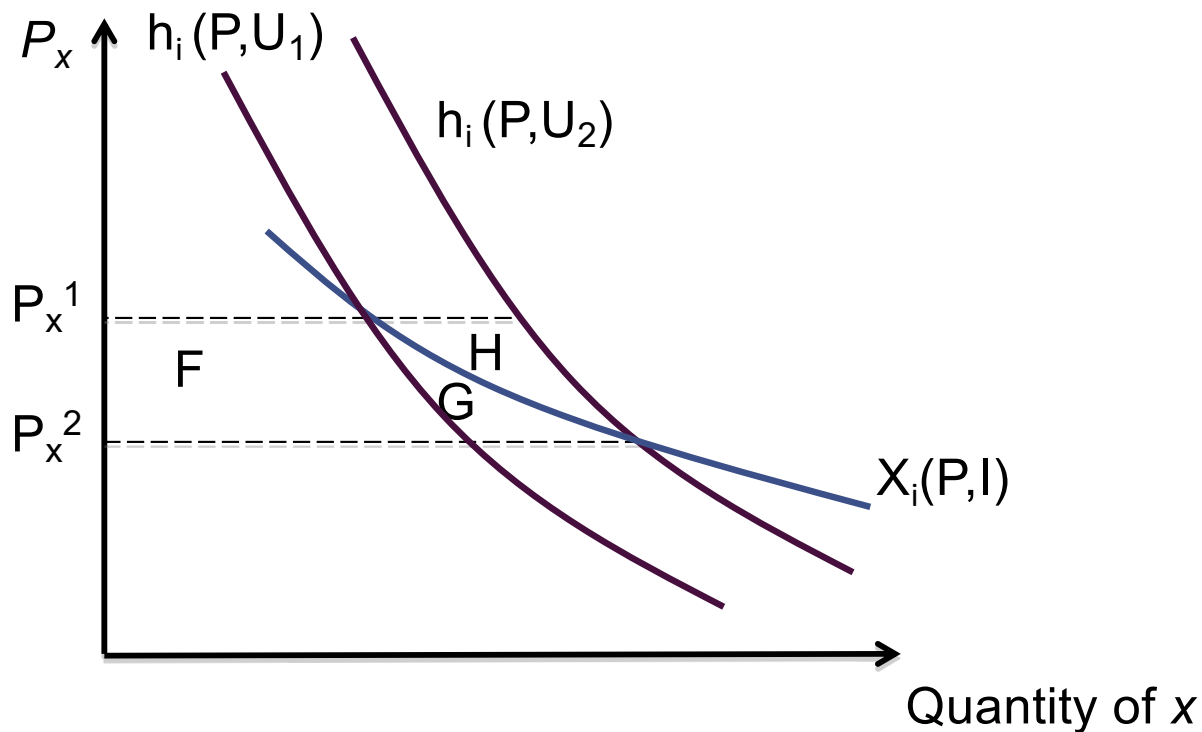
$$CS = - \int_1^4 \frac{I}{2P_x} dP_x = - \frac{8}{2} \ln P_x \Big|_1^4$$

$$CS = -4 \ln(4) + 4 \ln(1) = -5,545$$



Activity 4

Imagine a new economic policy that makes P_x decrease from P_x^1 to P_x^2 .



Indicate which areas represent CV? EV? And CS?

Activity 2: Solution

From last lecture, we calculated the Expenditure Function for this problem:

$$E(U, P_x, P_y) = 2U \sqrt{P_x P_y}$$

At point B: $E(U, P_x, P_y) = 2 \cdot 1 \sqrt{4 \cdot 4} = 8$

At point C: $E(U, P_x, P_y) = 2 \cdot 1 \sqrt{1 \cdot 4} = 4$

In this case, the individual will have to **give away \$4** of income, so $EV = -4$.

Consumer Welfare

Which of these measures should we use?

Economists use CV or EV to compute the welfare effects of price changes. Their definition allows the variation of more than one price.

When **only one price changes**, the Marshallian demand curve conveniently compromises these two measures. Hence, this is the measure of welfare losses we will primarily use.

Depending on the example, it may be attractive to use CV or EV:

- CV: if we are trying to arrange for some compensating scheme at the new prices
- If we are trying to get a reasonable measure of willingness to pay, EV is better because it measures the income change at current prices (it is easier for decision makers to judge the value of the dollar at current prices than at hypothetical prices). If we compare more than ONE policy change, the CV uses different base prices for each new policy while the EV keeps the same base price.

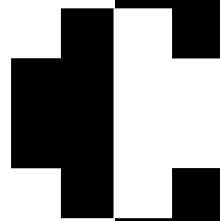
Consumer Theory

- Preferences and utility
- Budget constraint
- Indirect utility function
- Marshallian and Hicksian demands
- Expenditure function
- Slutsky equation
- Consumer surplus and individual welfare
- Aggregated demand

- **Production function**
- **Returns to scale**
- **Elasticity of substitution**
- **Isocost curves**
- **Minimization of costs**
- **Cost function**
- **Short- and long-run**
- **Producer surplus**
- **Aggregated supply**

Producer Theory

**Consumer
Theory**



**Producer
Theory**

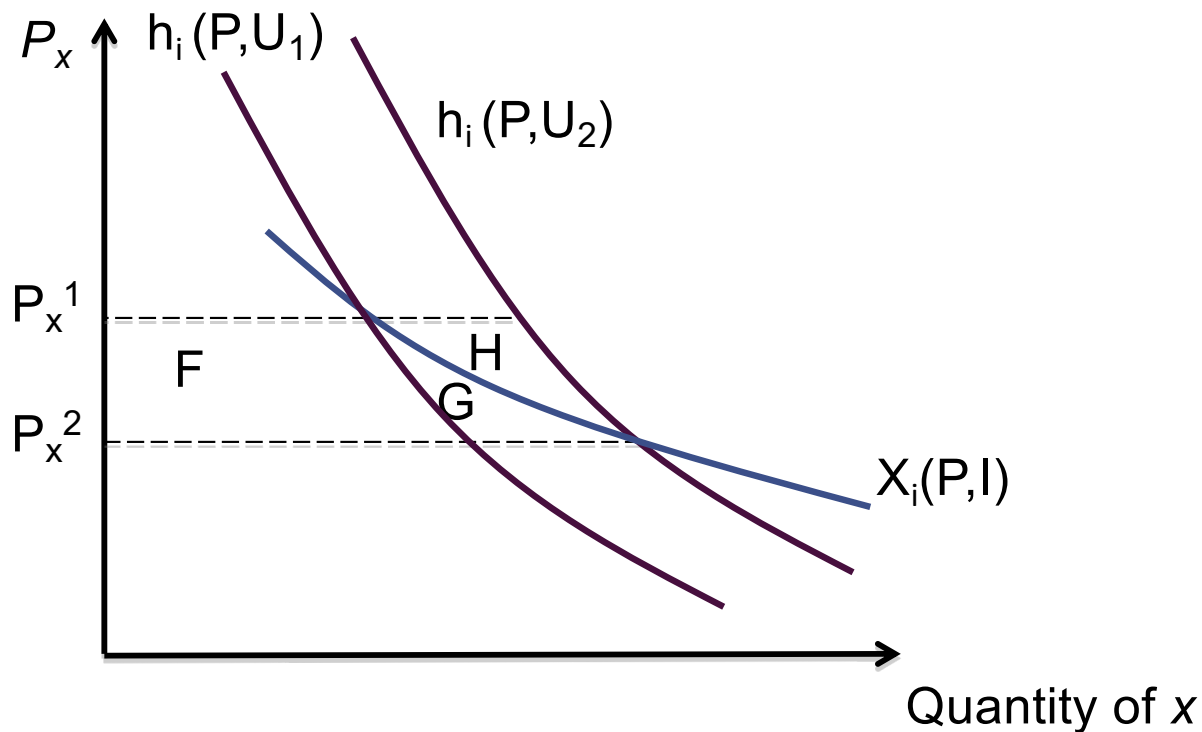
Welfare Economics and Competitive Markets

Market Failures

Game Theory

Activity 4: Solution

Imagine a new economic policy that makes P_x decrease from P_x^1 to P_x^2 .



$CV = F$, EV is $F + G + H$ and CS is $F + G$

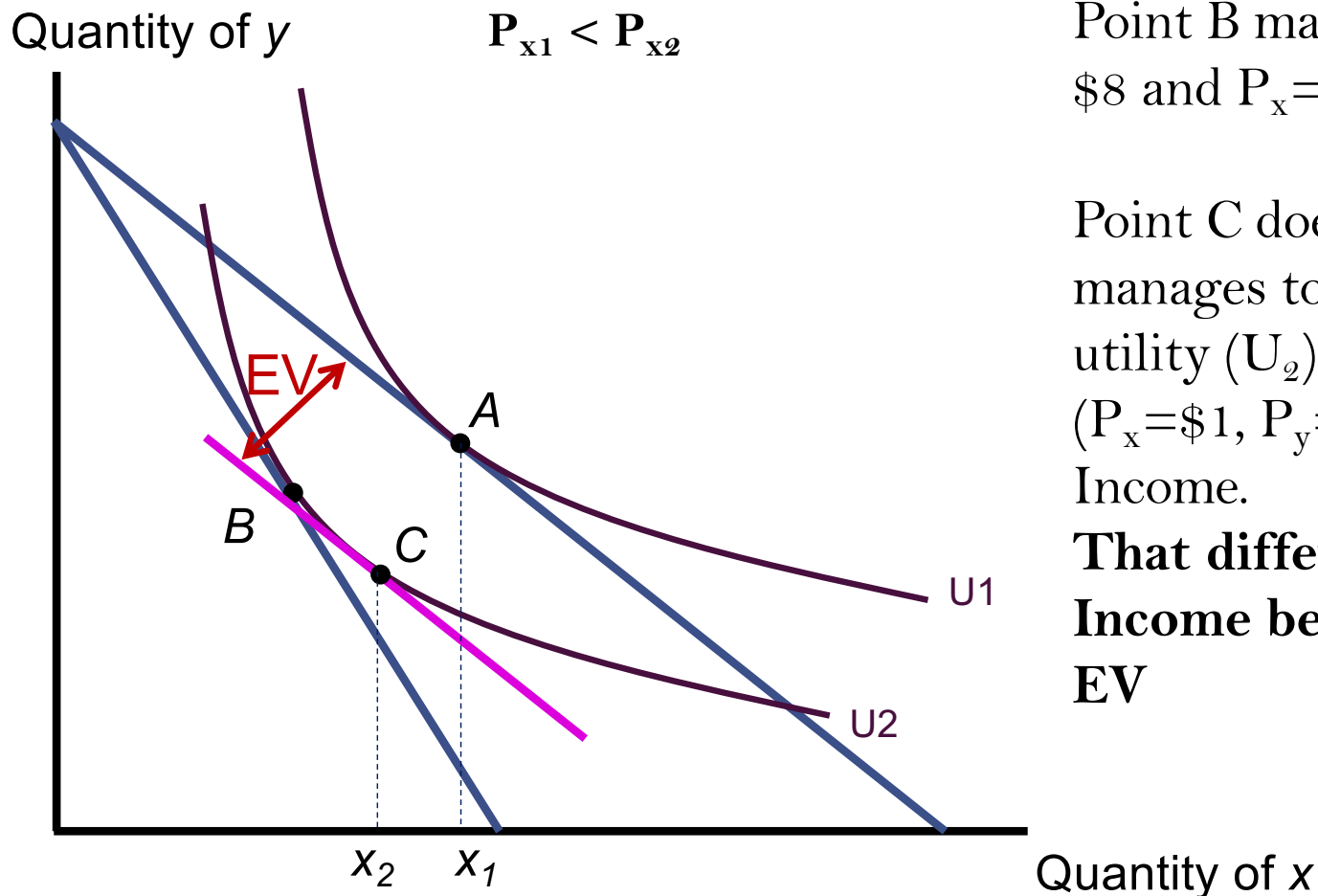
Activity 2: Solution

Point A maximizes utility at income \$8 and $P_x = \$1$ and $P_y = \$4$.

Point B maximizes utility at income \$8 and $P_x = \$4$ and $P_y = \$4$.

Point C doesn't maximize utility. It manages to reach the new level of utility (U_2), with the old set of prices ($P_x = \$1$, $P_y = \$4$) at a different level of Income.

That difference in the levels of Income between points B and C is EV



Individual Activity

In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.